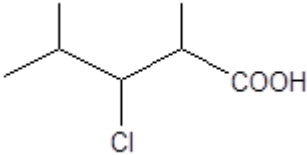
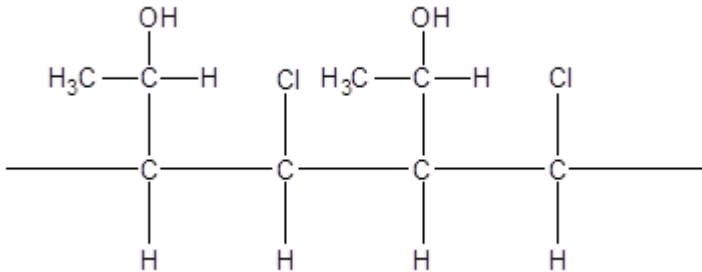


Section A

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
1.	(a)			3-chloro-2,4-dimethylpentanoic acid		1		1		
	(b)					1		1		
2.				Sketch graph to show products at higher energy than reactants (1) E_a and ΔH correct (1)		2		2		
3.						1		1		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4.				Any two of following for (1) each - Use a colorimeter to measure colour changes <u>with time</u> - Use a pH meter to measure pH changes <u>with time</u> - Use a gas syringe / collect gas over water to measure changes in volume <u>with time</u> - Use a balance to measure mass changes <u>with time</u> Award (1) for one correct method but no mention of time		2		2		2
5.	(a)			Presence of a (carbon to carbon) double bond	1			1		
	(b)			No – because two of the groups on one of the double bonded carbon atoms are the same		1		1		
6.				$\text{CuO} + 2\text{HCOOH} \rightarrow (\text{HCOO})_2\text{Cu} + \text{H}_2\text{O}$ (ignore state symbols)		1		1		
Section A total					1	9	0	10	0	2